Automated Market Makers: Mean-Variance Analysis of LPs Payoffs and Design of Pricing Functions

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Introduction

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Nice example of cross-learning between TradFi and DeFi

Classical (TradFi) trading mechanims

Different mechanisms to trade / take liquidity

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AMMs propose a new mechanism by which traders can swap one asset against another using a pool of reserves posted by other agents in quest of yield:

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- LPs supply reserves to the pool, and LTs use the AMM to trade

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Automated Market Makers

AMMs propose a new mechanism by which traders can swap one asset against another using a pool of reserves posted by other agents in quest of yield:

- LPs supply reserves to the pool, and LTs use the AMM to trade
- The exchange rate is based on a pre-defined and public rule / formula that depends on the reserves, the transaction size and the direction of the swap (and possibly other parameters) $\implies$ pricing function

Analysis

Main remarks

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- order-driven markets organized around limit order books $\implies$ in an AMM, prices are automatically set by the protocol.
- dark pools $\implies$ in an AMM, the available liquidity is visible (and the price is not defined solely by importing that of another venue)

Moreover, AMMs are fully collateralized and neither LPs nor LTs are exposed to any sort of counterparty risk.

The classical design of AMMs: CFMMs

Constant Function Market Makers

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- Exchange rate function of the reserves in the liquidity pool and the size of the transaction to keep a quantity constant:

$$f(q_t^0 - \Delta q^0, q_t^1 + \Delta q^1) = f(q_t^0, q_t^1)$$

i.e. keep $q_t^0 = \psi\left(q_t^1\right)$ all the time.

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CPMMs

The most common CFMMs are Constant Product Market Makers, where

$$f(q^0, q^1) = q^0 q^1$$

Setup of CFMMs

Exchange rate

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- if a client wants to sell Δq^1 coins of currency 1, she will receive Δq^0 coins of currency 0 such that

$$q_t^0 - \Delta q^0 = \psi\left(q_t^1 + \Delta q^1\right) \implies \frac{\Delta q^0}{\Delta q^1} = -\frac{\psi\left(q_t^1 + \Delta q^1\right) - \psi\left(q_t^1\right)}{\Delta q^1}$$

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- In an idealized world, there is an arbitrage if the market exchange rate S_t differs from $-\psi'\left(q_t^1\right)$

Analysis of LPs payoff in CFMMs

Fenchel-Legendre transform of ψ

$$\psi^* : p \in \mathbb{R}_-^* \mapsto \sup_{q \in \mathbb{R}_+^*} pq - \psi(q)$$

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- $\mathrm{PnL}_t - \mathrm{PnL}_t^{\mathrm{Hodl}} = \psi^*(-S_0) - \psi^*(-S_t) - (S_t - S_0)\psi^{*'}(-S_0) \leq 0$
  $\implies$ impermanent loss + option payoff ($\Gamma < 0$).

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In real CFMMs, transaction costs must "compensate" a short position in a convex option...

Analysis of LPs payoff in CFMMs

Itô's formula on the payoff

If ψ^* is twice continuously differentiable we have

$$\mathrm{PnL}_t - \mathrm{PnL}_t^{\mathrm{Hodl}} = \int_0^t (\psi^{*'}(-S_s) - \psi^{*'}(-S_0))dS_s - \frac{1}{2}\int_0^t \psi^{*''}(-S_s)d\langle S\rangle_s$$

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Hedging

The first term can be (theoretically) hedged away through a self-financing strategy... but this remains theoretical:

- At which frequency? On which venue(s)?
- How to proceed for an AMM?
- Is it desirable for an LP in several AMMs?
- What about feedback effects?

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LVR and price discovery

The second term is nonpositive. It is the price to pay to let LTs carry out price discovery.

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- Build in this paper a simple mean-variance framework inspired from the so-called modern portfolio theory, in order to compare different AMMs
- Estimate the maximum extra return that a LP could expect for a given level of tracking error with respect to Hodl
  $\implies$ efficient market making strategies

Efficient pricing functions in the perfect information case

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Exogenous market exchange rate

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

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Example

If a client wants to sell z coins of currency 0 at time t, then z/S_t coins of currency 1 will be offered to her and $z\delta^{0,1}(t,z)$ extra coins of currency 0 will be asked as a markup (in the model only... reality can be slightly different)

The model

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Markups process (spread capture)

$$dX_t = \int_{z \in \mathbb{R}_+^*} z\delta^{0,1}(t,z) J^{0,1}(dt,dz) + \int_{z \in \mathbb{R}_+^*} z\delta^{1,0}(t,z) J^{1,0}(dt,dz)$$

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Reserves

$$dq_t^0 = \int_{z\in\mathbb{R}_+^*} z\Big(J^{0,1}(dt,dz) - J^{1,0}(dt,dz)\Big) \text{ and } dq_t^1 = \int_{z\in\mathbb{R}_+^*} \frac{z}{S_t}\Big(J^{1,0}(dt,dz) - J^{0,1}(dt,dz)\Big)$$

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Transaction processes

Processes $J^{0,1}(dt,dz)$ and $J^{1,0}(dt,dz)$ with intensities

$$\nu_t^{0,1}(dz) = \Lambda^{0,1}\left(z, \delta^{0,1}(t,z)\right) \mathbb{1}_{\{q_{t-}^1 \geq \frac{z}{S_t}\}} m(dz)$$

and

$$\nu_t^{1,0}(dz) = \Lambda^{1,0}\left(z, \delta^{1,0}(t,z)\right) \mathbb{1}_{\{q_{t-}^0 \geq z\}} m(dz),$$

Comparison with Hodl

Excess reserves

We introduce the following two processes

$$\left(Y_t^0\right)_{t\in\mathbb{R}_+} = \left((q_t^0 - q_0^0)\right)_{t\in\mathbb{R}_+} \quad \text{and} \quad \left(Y_t^1\right)_{t\in\mathbb{R}_+} = \left((q_t^1 - q_0^1)S_t\right)_{t\in\mathbb{R}_+}$$

with dynamics

$$dY_t^0 = \int_{z\in\mathbb{R}_+^*} z\left(J^{0,1}(dt,dz) - J^{1,0}(dt,dz)\right)$$

and

$$dY_t^1 = \mu Y_t^1 dt + \sigma Y_t^1 dW_t + \int_{z\in\mathbb{R}_+^*} z\left(J^{1,0}(dt,dz) - J^{0,1}(dt,dz)\right)$$

Comparison with Hodl

Excess PnL

$$
\begin{aligned}
\mathrm{PnL}_T - \mathrm{PnL}_T^{\mathrm{Hodl}} &= X_T + Y_T^0 + Y_T^1 \\
&= \int_0^T \int_{z \in \mathbb{R}_+^*} z \delta^{0,1}(t,z) J^{0,1}(dt,dz) + \int_0^T \int_{z \in \mathbb{R}_+^*} z \delta^{1,0}(t,z) J^{1,0}(dt,dz) \\
&\quad + \int_0^T \mu Y_t^1 dt + \int_0^T \sigma Y_t^1 dW_t
\end{aligned}
$$

An optimization problem

Objective function

$$\sup_{(\delta^{0,1},\delta^{1,0})\in\mathcal{A}} \mathbb{E}\Bigg[\int_0^T \Bigg\{ \int_{z\in\mathbb{R}_+^*} \Big(z\delta^{0,1}(t,z)\Lambda^{0,1}(z,\delta^{0,1}(t,z))\mathbb{1}_{\{q_{t-}^1 \geq \frac{z}{S_t}\}}$$

$$+ z\delta^{1,0}(t,z)\Lambda^{1,0}(z,\delta^{1,0}(t,z))\mathbb{1}_{\{q_{t-}^0 \geq z\}}\Big) m(dz) + \mu Y_t^1 - \frac{\gamma}{2}\sigma^2 (Y_t^1)^2 \Bigg\} dt\Bigg]$$

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Remark

4 state variables $\implies$ numerically intractable

An optimization problem

Approximation

$$\sup_{(\delta^{0,1},\delta^{1,0})\in\mathcal{A}} \mathbb{E}\Bigg[\int_0^T \Bigg\{ \int_{z\in\mathbb{R}_+^*} \Big(z\delta^{0,1}(t,z)\Lambda^{0,1}(z,\delta^{0,1}(t,z))$$

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Remark

Only one state variable, Y^1, with Markovian dynamics $\implies$ tractable

An optimization problem

HJB equation

$$\begin{cases} 0 = \partial_t \theta(t,y) + \mu y(1+\partial_y \theta(t,y)) - \frac{\gamma}{2}\sigma^2 y^2 + \frac{1}{2}\sigma^2 y^2 \partial^2_{yy}\theta(t,y) \\ \qquad + \int_{\mathbb{R}_+^*} \left(zH^{0,1}\left(z, \frac{\theta(t,y)-\theta(t,y-z)}{z}\right) + zH^{1,0}\left(z, \frac{\theta(t,y)-\theta(t,y+z)}{z}\right)\right) m(dz) \\ \theta(T,y) = 0 \end{cases}$$

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Hamiltonian functions

$$H^{0,1}(z,p) = \sup_{\delta \geq -C} \Lambda^{0,1}(z,\delta)(\delta - p) \text{ and } H^{1,0}(z,p) = \sup_{\delta \geq -C} \Lambda^{1,0}(z,\delta)(\delta - p)$$

An optimization problem

Optimal controls

The supremum in the definition of $H^{i,j}(z,p)$ is reached at a unique $\bar{\delta}^{i,j}(z,p)$ given by

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The markups that maximize our modified objective function are obtained in the following form

$$\delta^{0,1*}(t,z) = \bar{\delta}^{0,1}\left(z, \frac{\theta(t, Y^1_{t-}) - \theta(t, Y^1_{t-} - z)}{z}\right)$$

and

$$\delta^{1,0*}(t,z) = \bar{\delta}^{1,0}\left(z, \frac{\theta(t, Y^1_{t-}) - \theta(t, Y^1_{t-} + z)}{z}\right).$$

A mean-variance analysis

Market simulator

A mean-variance analysis

Market simulator

- Currency 0: USD, Currency 1: ETH

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Market simulator

- Currency 0: USD, Currency 1: ETH
- Initial exchange rate: 1600 USD per ETH

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- Time horizon: $T = 0.5$ day

Efficient frontier

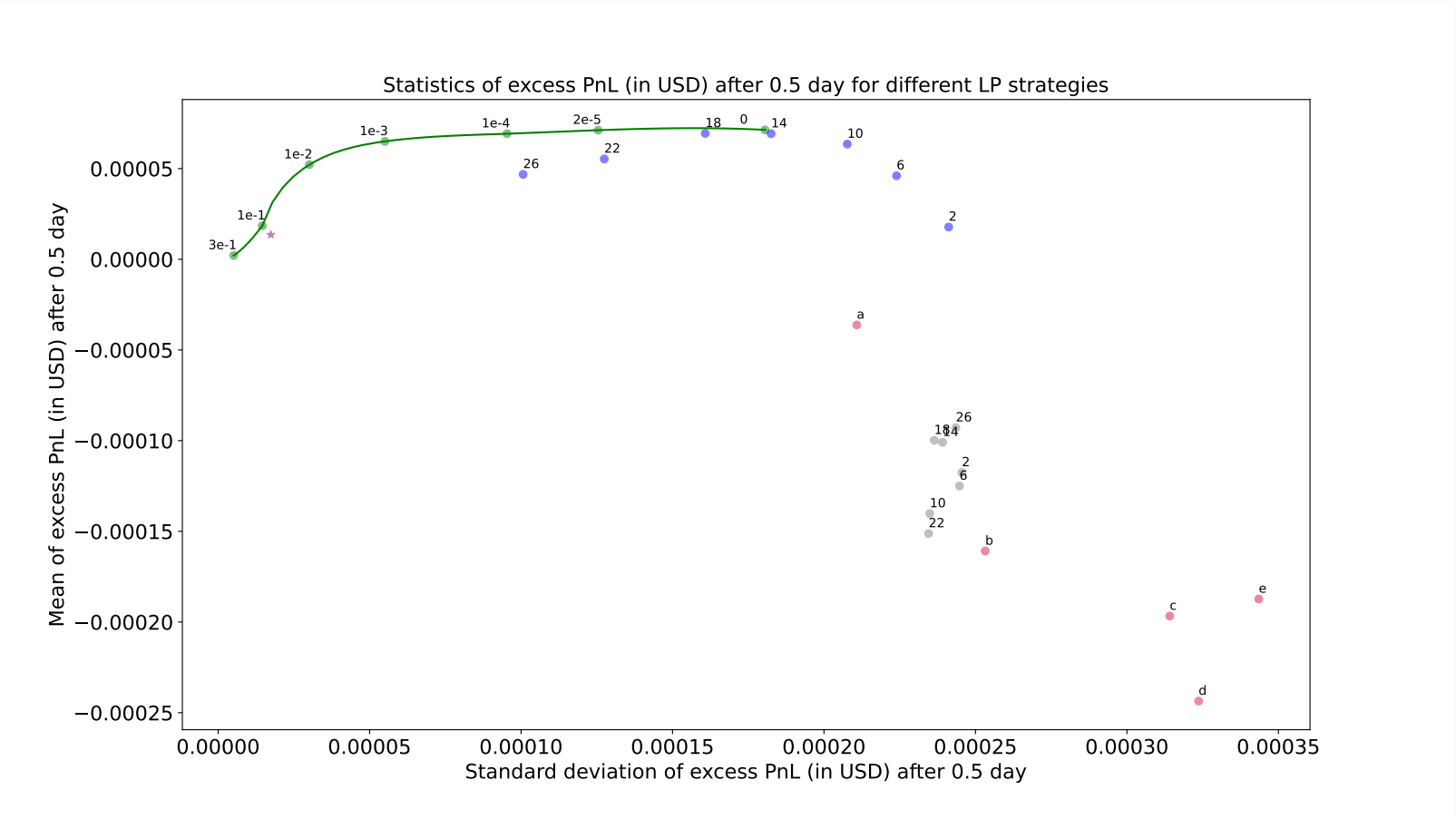

Figure 1: In blue: naive strategies with constant $\delta^{0,1}, \delta^{1,0}$. In grey: CPMM with fees. In pink: other CFMM without market exchange rate, for different sets of realistic parameters. In purple ($\star$): CFMM with market exchange rate oracle. In green: efficient frontier, obtained using the optimal markups for different levels of risk aversion.

Misspecification, partial information and arbitrages

Misspecification

On $\lambda^{i,j}$ and σ

Assuming $\mu = 0$, our control problem can be written as

$$\sup_{(\delta^{0,1},\delta^{1,0})\in\mathcal{A}} \mathbb{E}\Bigg[\int_0^T \Bigg\{ \int_{z\in\mathbb{R}_+^*} \Bigg(z\delta^{0,1}(t,z)\frac{\Lambda^{0,1}(z,\delta^{0,1}(t,z))}{\gamma\sigma^2}$$

$$+z\delta^{1,0}(t,z)\frac{\Lambda^{1,0}(z,\delta^{1,0}(t,z))}{\gamma\sigma^2}\Bigg) m(dz) - \frac{1}{2}(Y_t^1)^2 \Bigg\} dt\Bigg]$$

Misspecification: $\lambda^{i,j}$

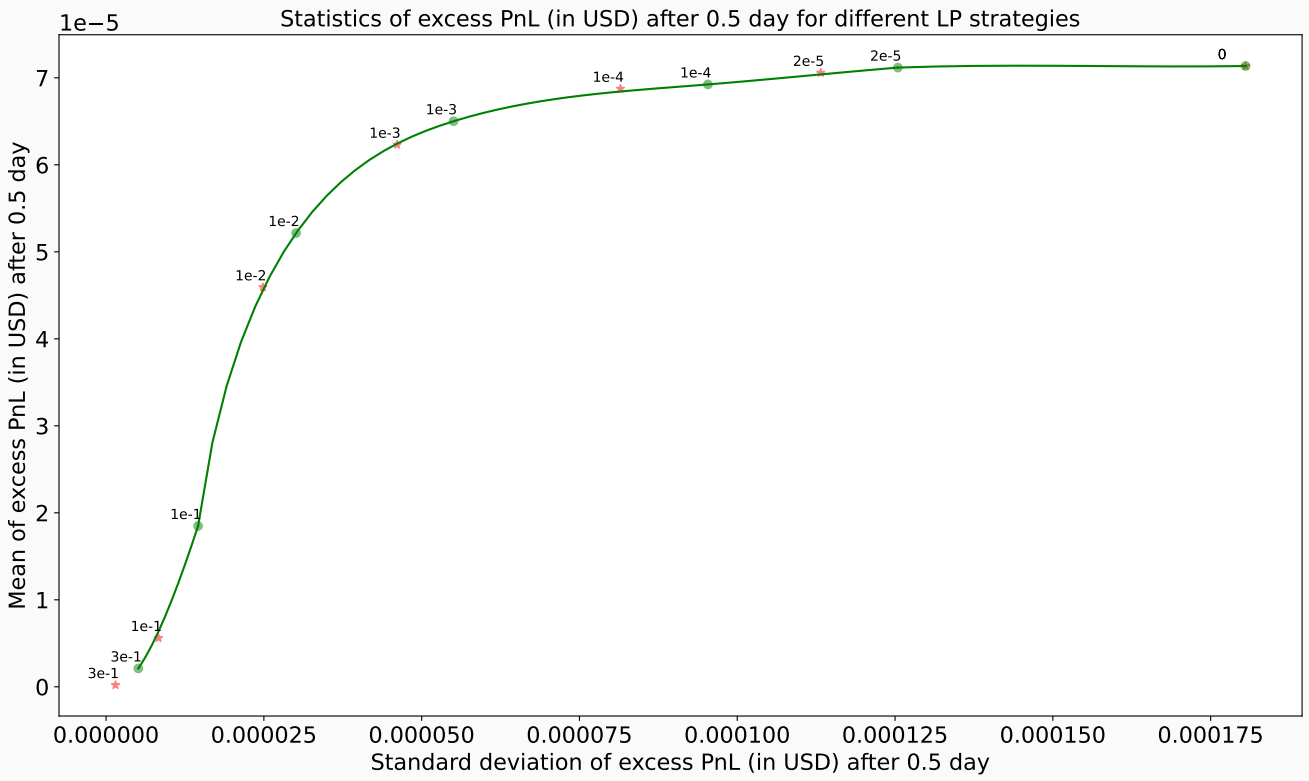

Figure 2: Performance of strategies in terms of mean / standard deviation of excess PnL when $\lambda^{0,1}$ and $\lambda^{1,0}$ are misspecified. In green: efficient frontier, obtained with the efficient strategy for different levels of risk aversion with complete information. In pink: performance of the misspecified strategy obtained with $\lambda^{0,1} = \lambda^{1,0} = 50$ day^{-1} for different levels of risk aversion.

Misspecification: σ

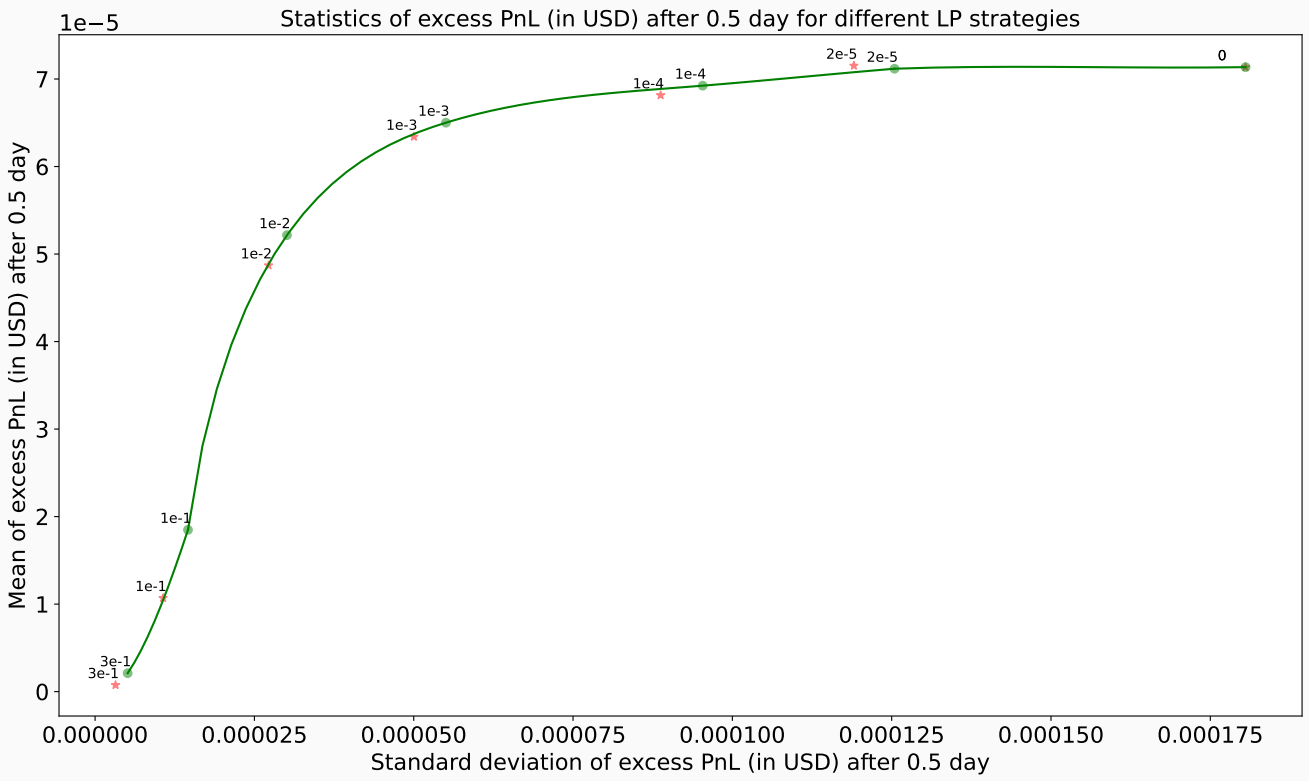

Figure 3: Performance of strategies in terms of mean / standard deviation of excess PnL when σ is misspecified. In green: efficient frontier, obtained with the efficient strategy for different levels of risk aversion with complete information. In pink: performance of the misspecified strategy obtained with $\sigma = 1.2$ year$^{-\frac{1}{2}}$ for different levels of risk aversion.

Partial information

Oracles

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Oracles

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- In practice, we can feed a smart contract with external data through an oracle, but this can only be done at discrete times.

Partial information

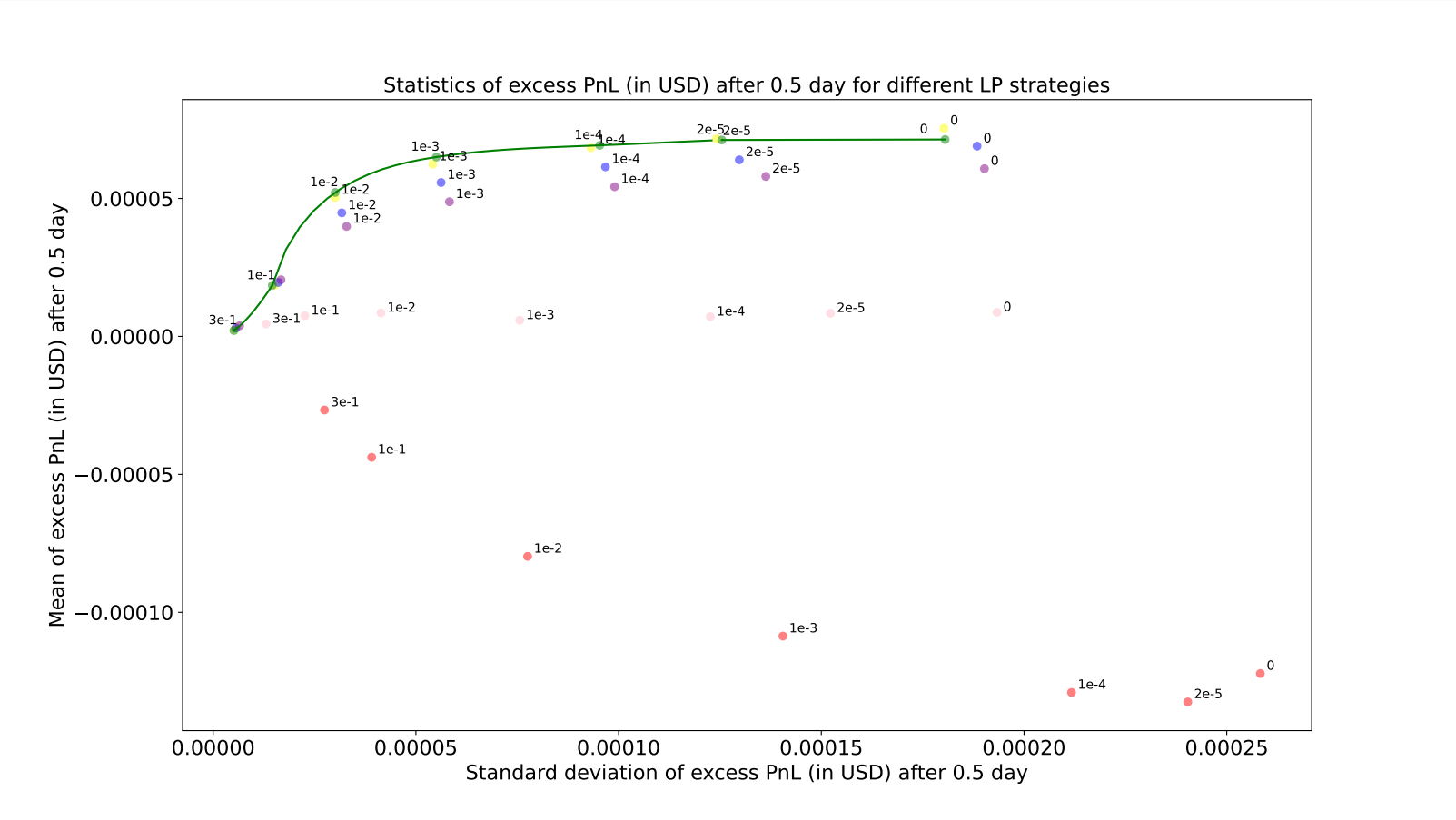

Figure 4: Performance of the efficient strategies in terms of mean / standard deviation of excess PnL, obtained by playing the efficient strategies for different levels of risk aversion with different oracle delays: complete information (in green), 10-second delay (in yellow), 30-second delay (in blue), 1-minute delay (in purple), 5-minute delay (in pink), 30-minute delay (in red).

Arbitrage

Systematic arbitrage

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Systematic arbitrage

- Partial information regarding the market exchange rate can sometimes result in arbitrage opportunities for LTs
- Already taken into account in the demand curves modeled by the intensity functions, though not in a systematic way: if a price appears to be very good for LTs, the probability that a transaction occurs is very high
- In practice however, there exists a category of agents, called arbitrageurs, who systematically exploit arbitrage opportunities: they trade with the AMM until arbitrage opportunities disappear

Arbitrages

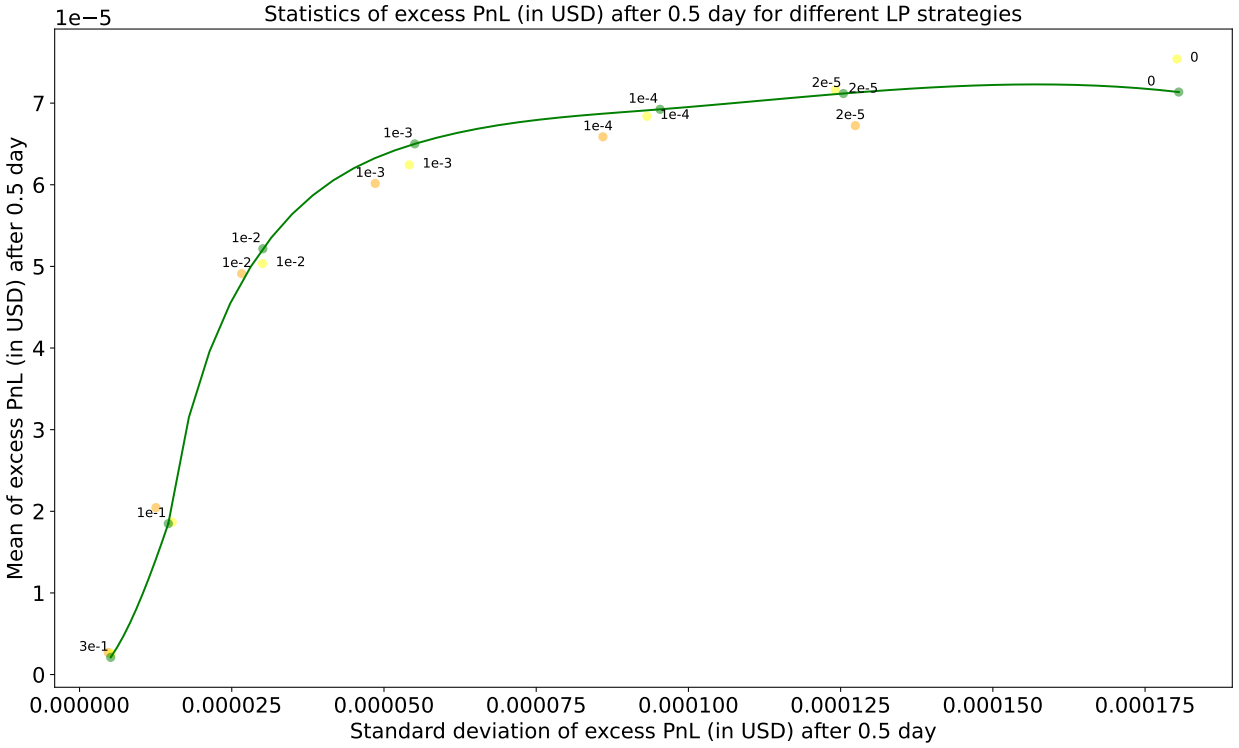

Figure 5: In green: efficient frontier, obtained by playing the optimal strategy for different levels of risk aversion with complete information. In yellow: performance of the same optimal strategy for different levels of risk aversion with a lagged oracle. In orange: performance of the same optimal strategy for different levels of risk aversion with a lagged oracle and with arbitrage flow.

Arbitrages

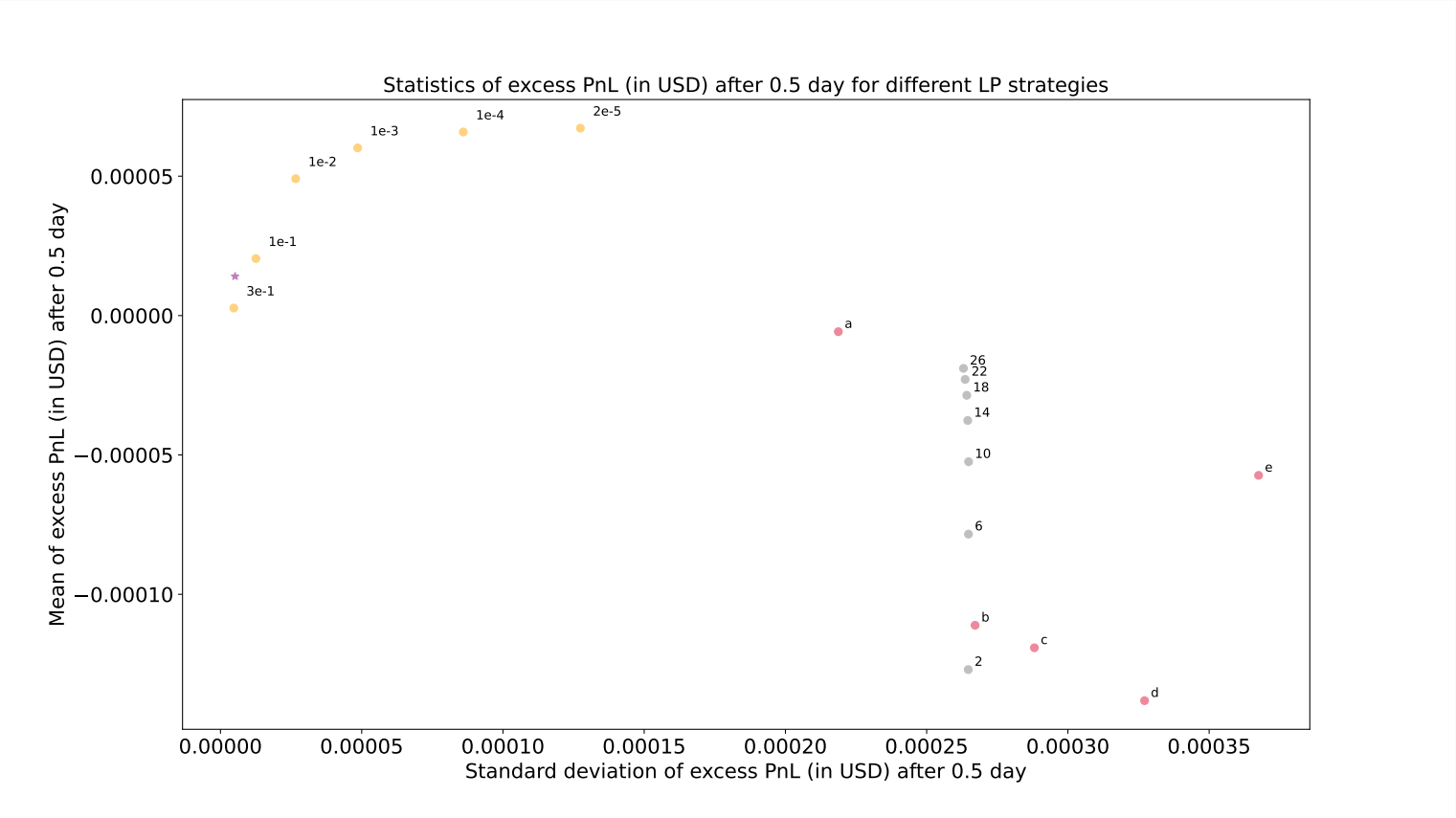

Figure 6: Performance with a lagged oracle and arbitrage flow. In grey: CPMM with fees. In pink: CFMM without market exchange rate oracle (like Curve) for different sets of realistic parameters. In purple ($\star$): AMM with market exchange rate oracle (like the first version of Swaap). In orange: performance of the efficient strategies for different levels of risk aversion.

Summary

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Main results

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- Allowing an AMM to get information about the current market exchange rate (through an oracle) can significantly improve performance.
  $\implies$ significantly reduces the volatility of the excess PnL while delivering a positive excess PnL on average.
- Introducing an oracle in the AMM design comes at the cost that the oracle itself should be carefully designed to avoid introducing additional attack vectors.

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A special issue of Applied Mathematical Finance (co-ed. with Agostino Capponi)

The Microstructure of Decentralized Finance

The End

Thank you. Questions?